

IDS Exemplar Lab Write-Up

AI + Cybersecurity Faculty Summer Institute

Topic 06 Afternoon Activity

Purpose

This exemplar lab shows how a small machine learning activity can teach security workflow design. Students work with flow-like network records, train a simple classifier, inspect metrics, and explain what the model output would mean for an intrusion detection workflow.

The goal is not to build a production IDS. The goal is to make the security task, data assumptions, model role, metrics, and human review point visible enough for classroom discussion.

Learning Goals

By the end of the lab, students should be able to:

- describe how network-flow features can support a simple intrusion detection task;
- train and evaluate a basic classifier on labeled security data;
- interpret false positives and false negatives as operational tradeoffs;
- explain what a human analyst would still need to review before acting on an alert.

Dataset

The starter dataset is a small synthetic flow table. Each row represents a connection summary rather than raw packets. The fields include:

- protocol, duration, bytes sent, bytes received, packet count, destination port, failed login count, and connection rate;
- a binary label indicating benign or suspicious activity.

The synthetic data is intentionally simple. It is appropriate for demonstrating workflow and tradeoffs, but it should not be presented as realistic network telemetry.

Student Task

Students complete a Colab notebook that asks them to:

1. load the flow dataset;

2. inspect class balance and feature distributions;
3. split the data into training and test sets;
4. train a simple classifier;
5. evaluate accuracy, precision, recall, and the confusion matrix;
6. adjust or discuss a decision threshold;
7. write a short interpretation of what the results mean for IDS use.

Discussion Questions

- Which errors are more costly in this lab: false positives or false negatives?
- What would a human analyst need to see before acting on a model alert?
- What assumptions does the synthetic dataset make about attack behavior?
- How would the lab change if students used real trace data instead of a small synthetic table?
- What part of this lab would transfer well to your classroom, and what would need redesign?

Live Demo Plan

The facilitator can demo only part of the notebook:

- load the dataset and inspect the label distribution;
- train the baseline classifier;
- show the confusion matrix;
- pause on one false positive and one false negative to discuss operational meaning.

Design Takeaway

A good security-ML lab should make the student answer more than “did the model perform well?” It should also make students ask what task the model supports, what evidence it uses, what errors cost, and where human judgment remains necessary.

Student-Facing Rubric

Category	Points	What strong work demonstrates
Security task framing	15	Clearly explains the IDS task and what the model output supports.
Data understanding	15	Identifies label balance, feature meaning, and synthetic-data limits.
Model implementation	20	Runs the classifier workflow and produces reproducible outputs.
Evaluation interpretation	20	Connects metrics and confusion-matrix results to security consequences.
Human review and limitations	15	Explains what an analyst still needs to inspect or decide.
Communication	15	Presents conclusions clearly and connects the lab to classroom use.